

Project_1_8420

March 3, 2025

1 DSA 8420 Spring 2025 Project 1

Due March 11, 2025

This project consists of two problems. In the first problem, we solve and analyze the agricultural problem presented in Homework 2. In the second problem, we examine a linear system with conflicting equations and compute a measure of its inconsistency.

```
[1]: # packages
import pyomo.environ as pyo
from pyomo.opt import SolverFactory
```

1.1 The Agricultural Problem

Problem description: A farmer owns a farm that produces corn, soybean, and oats. There are 12 acres of land available for cultivation. Each crop that is planted has certain requirements for labor and capital. These data along with the net profit figure are given in the accompanying table.

	<i>Labor (hr)</i>	<i>Capital (\$)</i>	<i>Net profit (\$)</i>
Corn (per acre)	6	36	40
Soybeans (per acre)	6	24	30
Oats (per acre)	2	18	20

```
[ ]:
```